

**University of BLIDA 1**  
Science Faculty  
Computer Science Department



## **FINAL REPORT**

Third year Engineer (Cyber Security)

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# **DEVELOPING AN INTEGRATED EDUCATIONAL PLATFORM FOR PRIVATE SCHOOLS: ACHIEVING DIGITAL TRANSFORMATION**

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# Abstract

"Schooline" is a comprehensive Arabic online platform designed to simplify the learning process for high school students. It also offers a variety of courses to develop their skills beyond the high school curriculum, such as programming, languages, and other useful skills. The platform provides a wide range of paid and free courses delivered by high-quality teachers, addressing common challenges faced by students, such as difficulty in choosing courses and selecting suitable teachers. Additionally, the platform offers special services for teachers, including professional support in registration and material uploading, personalized assistance, marketing support, and flexible preparation options, thereby fostering an effective learning and teaching environment.

## **KEYWORDS**

comprehensive Arabic Online platform, high school students, various courses, high-quality instructors, special services, conducive environment.

# Résumé

« Schooline » est une plateforme arabe en ligne complète conçue pour simplifier le processus d'apprentissage des élèves du secondaire. En plus, elle offre divers cours pour développer leurs compétences en dehors du cadre scolaire, tels que l'apprentissage de la programmation, des langues et d'autres compétences utiles. La plateforme propose une large gamme de cours payants et gratuits dispensés par des enseignants de haute qualité, répondant ainsi aux défis courants auxquels les étudiants sont confrontés, comme la difficulté de choisir des cours et de sélectionner les enseignants appropriés. De plus, la plateforme offre des services spéciaux pour les enseignants, y compris un soutien professionnel pour l'inscription et le téléchargement des matériaux, une assistance personnalisée, un soutien marketing et des options de préparation flexibles, favorisant ainsi un environnement d'apprentissage et d'enseignement efficace.

## **MOTS CLÉS**

Plateforme arabe en ligne complète, élèves du secondaire, divers cours, enseignants de haute qualité, services spéciaux, environnement propice.

## ملخص

« Schooline » هي منصة عربية عبر الإنترنت شاملة مصممة لتبسيط عملية التعلم لطلاب المدارس الثانوية، بالإضافة إلى تقديم دورات متنوعة لتطوير مهاراتهم خارج إطار الثانوية مثل تعلم البرمجة واللغات وغيرها من المهارات المفيدة. تقدم المنصة مجموعة واسعة من الدورات المدفوعة والمجانية التي يقدمها أساتذة ذوي جودة عالية، مما يعالج التحديات الشائعة التي تواجه الطلاب، مثل صعوبة اختيار الدورات واختيار المعلمين المناسبين. بالإضافة إلى ذلك، توفر المنصة خدمات خاصة للمعلمين، بما في ذلك الدعم المهني في التسجيل وتحميل المواد، والمساعدة الشخصية، والدعم التسويقي، وخيارات التجهيز المرنة، مما يعزز بيئة ملائمة لعملية التعلم والتدريس بشكل فعال.

### الكلمات المفتاحية

منصة عربية عبر الإنترنت، طلاب المدارس الثانوية، دورات متنوعة، خدمات خاصة، أساتذة ذوي جودة عالية، بيئة ملائمة.

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# 1 INTRODUCTION

Private schools face numerous obstacles that hinder their educational efficiency, ranging from funding shortages and competitive landscapes to limited salaries and academic pressure. Upon closer examination of the students applying to these schools, it has become evident that most of them struggle to keep up with their academic lives, experiencing wasted efforts that lead to fatigue and hinder their ability to perform at their best. Hence, the idea for our project, "Schooline", emerged.

"Edu Web" is an integrated educational platform designed to streamline the learning process and enhance the quality of education in private schools. It achieves this by offering a wide range of courses at affordable prices for secondary education across all academic streams. Additionally, it provides diverse and valuable courses in areas such as language learning, design, programming, and other sciences suitable for individuals of all backgrounds.

Our project offers several solutions. Firstly, we have developed an educational interface that enables students to find suitable teachers through a comprehensive teacher and course evaluation system, which showcases each teacher's popularity and effectiveness. We have also transitioned traditional teaching methods into digital ones, which brings multiple benefits such as reduced financial expenses and improved work quality. This is accomplished by providing modern teaching technologies and developing training programs for faculty members to enhance their teaching skills.

Moreover, we facilitate time and effort savings for teachers by recording lessons with the assistance of website personnel and storing them on the platform for students to access at their convenience. This creates a comfortable and encouraging work environment free from distractions, thereby preventing students from experiencing chaos and stress, and ultimately enhancing productivity and fostering a better educational environment.

The remaining of this report is structured in three sections following the "3C" approach: Conceptualize, Conduct, and Conclude.

## **Section 1:** Understanding the problem and analyzing the current situation

We focused in this section on studying the core idea of the project from various angles and identifying obstacles that our project may encounter, then devising solutions for them. We began documenting implementation tools.

**Section 2:** Designing the solution

In this section, we focused on the detailed design of the project, including specifying requirements, creating various diagrams, designing the database schema, and the website logo.

**Section 3:** Project execution

In this third and final section, we designed the interfaces and implemented them with their back-end functionalities, linking them together to complete the website.

## 2 DESIGN

The design and analysis phase are a fundamental step in the development of a system, helping to create an effective website that achieves its business objectives and provides a productive experience for users. It involves detailing various diagrams and scenarios to be implemented in the website's development phase, which will help better understand the website development using the UML language.

### 2.1 Identification of Actors

The actors who will intervene in our system are:

- Student: create account, buy course, follow teacher, post comment, send feedback, transfer credits.
- Teacher: all his interactions are with the tech team; he just sends the meeting posts.
- Admin: adding rules to the website.
- Tech team: handling website-related issues.

### 2.2 Use case diagram

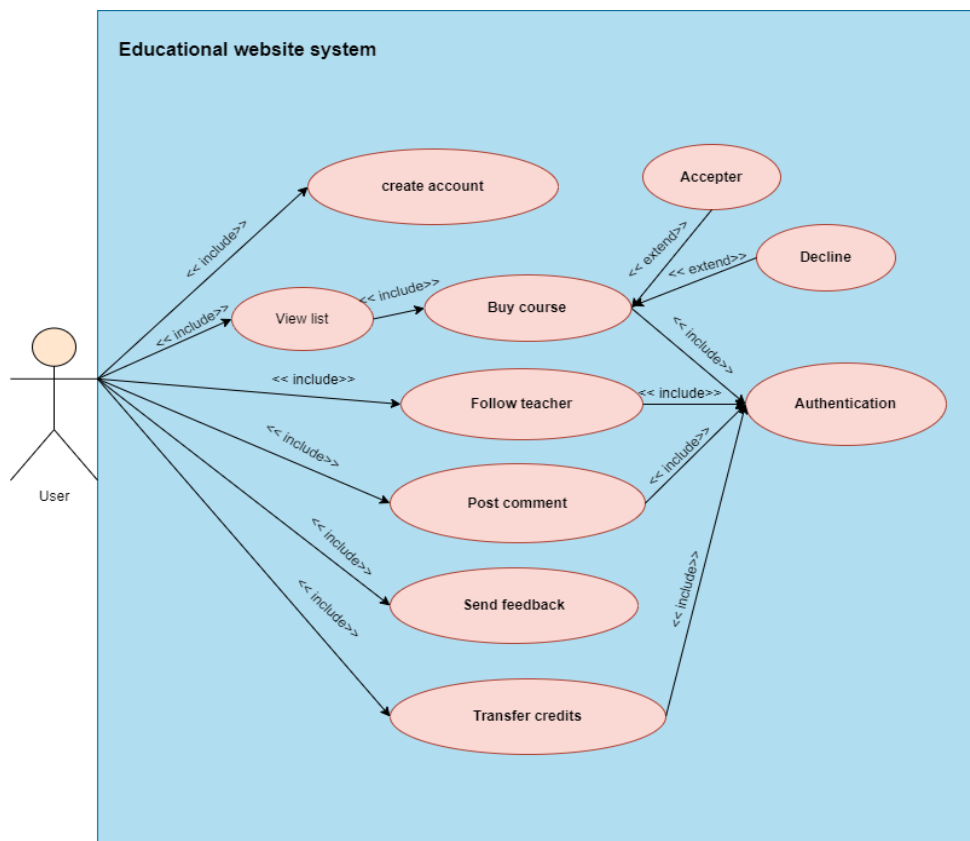


Figure 1.1 : Use case diagram for user.



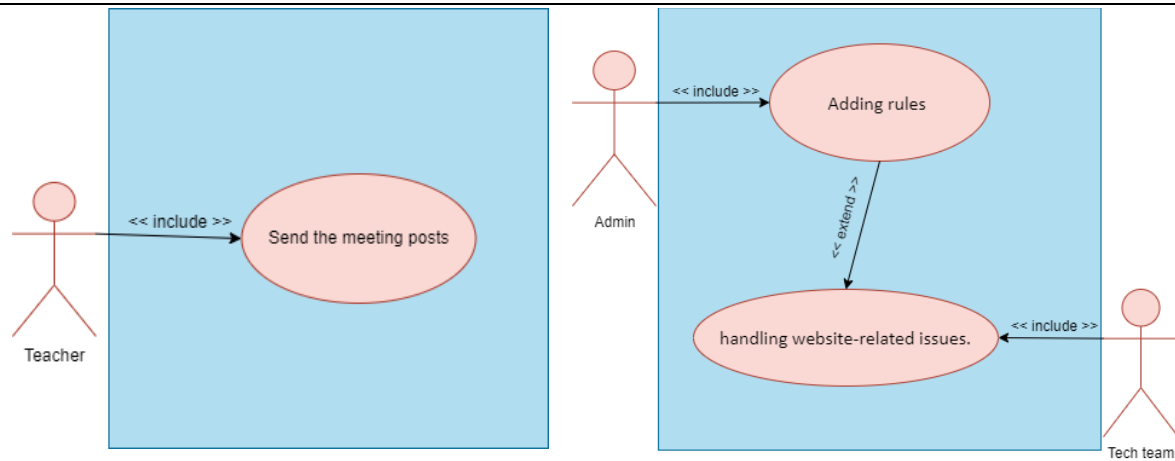


Figure 1.2 : Use case diagram for teacher, admin & tech team.

## 2.3 UML

The Unified Modeling Language (UML) is a standard graphical modeling language consisting of diagrams used to visualize and describe the structure and behavior of objects within a system. It is utilized to enhance communication and understanding of complex computer systems among developers.

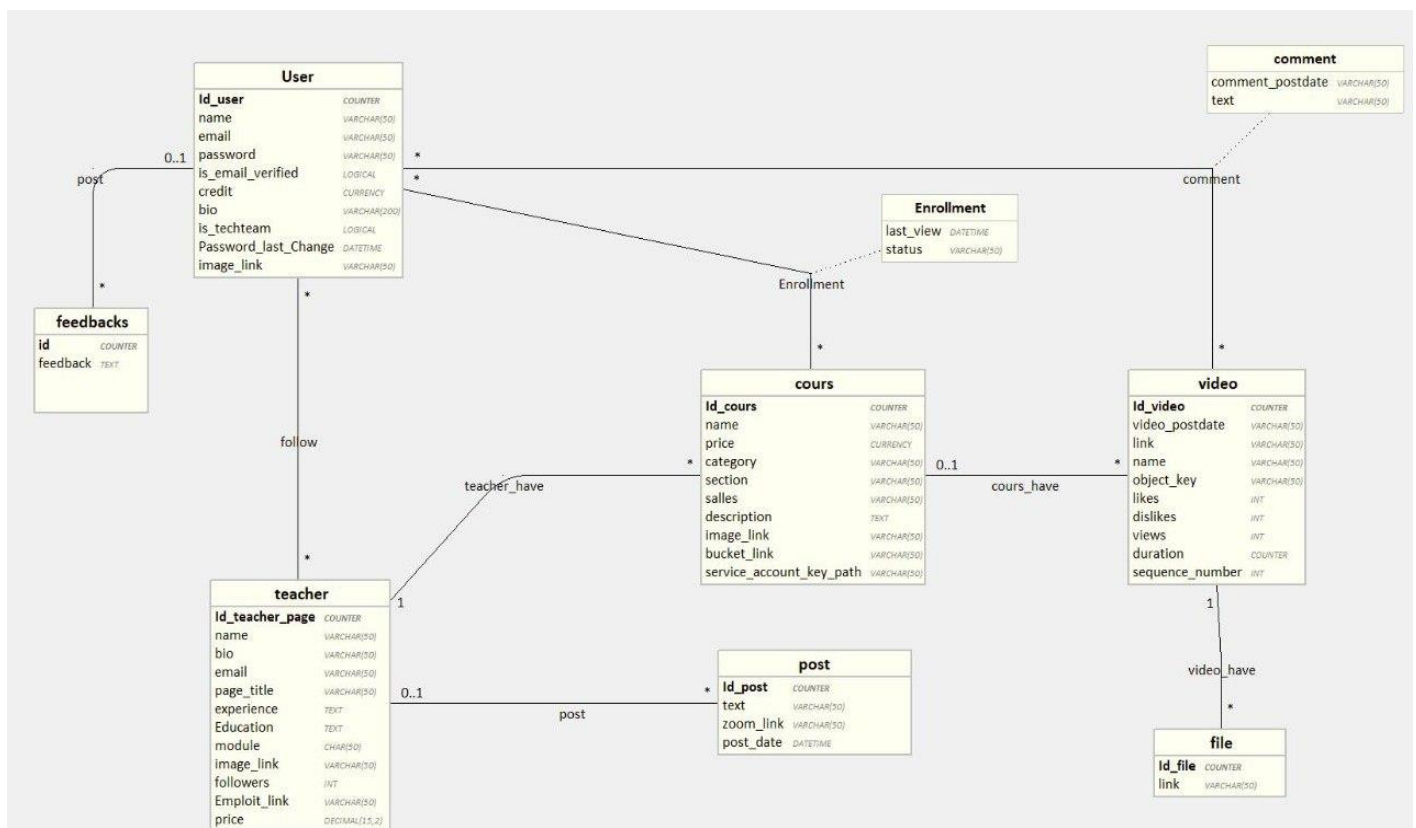


Figure 1.3 : The UML diagram of "Schooline".

## 3 IMPLEMENTATION

Implementation is the process of defining and implementing a conceptual plan to provide users with a visual interface to the work. In this chapter, we will introduce the relational model, the different tools and software used to develop this website, and finally, the different screenshots obtained from testing this web application as they are visible to the user.

### 3.1 RELATIONAL MODEL

**Teacher** (Id\_teacher\_page COUNTER, name VARCHAR (50), bio VARCHAR (50), email VARCHAR (50), page\_title VARCHAR (50), experience TEXT, Education TEXT, module CHAR (50), image\_link VARCHAR (50), followers INT, Emplot\_link VARCHAR (50), price DECIMAL (15,2)).

**Cours** (Id\_cours COUNTER, name VARCHAR (50), price CURRENCY, category VARCHAR (50), section VARCHAR (50), salles VARCHAR (50), description TEXT, image\_link VARCHAR (50), bucket\_link VARCHAR (50), service\_account\_key\_path VARCHAR (50), #Id\_teacher\_page).

**Student** (Id\_student COUNTER, name VARCHAR (50), email VARCHAR (50), password VARCHAR (50)).

**Post** (Id\_post COUNTER, text VARCHAR (50), zoom\_link VARCHAR (50), post\_date DATETIME, #Id\_teacher\_page).

**video** (Id\_video COUNTER, video\_postdate VARCHAR (50), link VARCHAR (50), name VARCHAR (50), object\_key VARCHAR (50), likes INT, dislikes INT, views INT, duration COUNTER, sequence\_number INT, #Id\_cours).

**file** (Id\_file COUNTER, link VARCHAR (50), #Id\_video).

**Comment** (Id\_comment COUNTER, comment\_postdate VARCHAR (50), text VARCHAR (50), #Id\_video, #Id\_user).

**Enrollment** (last\_view DATETIME, status VARCHAR (50), #Id\_cours, #Id\_user).

**follow** (#Id\_user, #Id\_teacher\_page).

**Feedback** (Id COUNTER, feedback TEXT).

### 3.2 Logo design

The vibrant color palette of our logo reflects the dynamic and engaging nature of the “Schooline” learning experience. The bold hues of blue, green, and yellow symbolize knowledge, growth, and optimism, respectively, perfectly encapsulating the essence of our educational philosophy.



Figure 1.4 : Schooline’s logo

The carefully crafted icons embedded within the logo serve as visual representations of the diverse course offerings available on “Schooline”. The laptop symbolizes our dedication to providing IT training, while the microscope and books represent our commitment to fostering academic excellence in a variety of subjects.

This logo was meticulously designed using Adobe Photoshop, meticulously crafted to embody the spirit of Schooline and serve as a beacon of inspiration for students seeking to expand their horizons and achieve their full potential.

### 3.3 Figma

Figma is a user interface design tool that allows for presenting the interface of a website, creating animations, and understanding client needs. It streamlines the work of developers by providing an overall view of the website's appearance.



### 3.4 Photoshop

Photoshop is a software that provides tools for making modifications to images used on a website, such as changing colors, adding or removing elements, and more.



### 3.5 Canva

Canva is an online graphic design tool that facilitates the creation of professional designs quickly. It can be used to create stickers, cards, logos, presentations, etc. Provides templates and editing tools for texts, photos, shapes, drawings and backgrounds.



### 3.6 VS Code

It is an open-source text editor and integrated development environment (IDE) developed by Microsoft, and it works on Windows, macOS, and Linux operating systems. VS Code supports a wide range of programming languages such as Python and JavaScript, and it provides a rich development environment with features like code editing, debugging, version control integration, and extensions for additional functionalities.



### 3.7 GitHub

is a web service hosting and development management of software, using Git version control software. GitHub offers accounts paid professionals, as well as free accounts for free software projects. The site also provides access control



and collaboration features such as bug tracking, feature requests, task management and a wiki for each project. The Git hub name consists of the word “git” referring to an open-source version control system and the word «hub» referring to the network social built around the Git system.

### 3.8 Frontend

#### 3.9 HTML (Hyper Text Markup Language):

It is a markup language used to build and organize the content of web pages. It consists of a set of elements and tags that define the structure and content of the page, and it can interact with other languages such as CSS and JavaScript to enhance and design the pages.



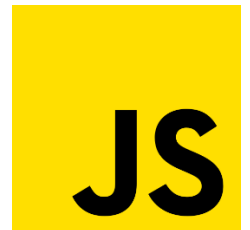
#### 3.10 CSS (Cascading Style Sheets):

It is a styling language used to define the visual appearance and formatting of HTML-written web pages. It allows control over elements such as colors, fonts, layout, styling, graphics, animations, and more.



#### 3.11 JavaScript:

It is a high-level, interpreted programming language used widely for building dynamic web applications and creating interactive user interfaces. JavaScript can be used to add interactivity, animated graphics, and effects to web pages.



### 3.12 Backend

#### 3.13 Django:

It is a high-level web framework written in Python that encourages rapid development, clean and pragmatic design. Django has several key features such as the Model-View-Template (MVT) architecture for separating application logic, a powerful ORM system for interacting with the database using Python, built-in admin interface for managing the application and users, a flexible template system for separating the presentation logic, advanced form handling for managing user input, built-in user authentication and authorization systems, comprehensive testing framework, and database migration system.

Our website is formed from many facades to provide the various services to be reached to achieve its objectives. Among these interfaces are:

### 3.14 Home Page

The home page contains a brief description of the idea of our completed site attached, as well as the latest courses offered by the school that must be logged in to be joined, and a list of our teachers responsible for offering these courses. The homepage also includes a section on our services for high school and other educational courses, which are links that display a special page of lessons, a section to develop comments and opinions about our platform with a view to further development, and a part of which is statistics on the number of pupils involved with us, the number of staff instructors and the hours of study to give comprehensive ideas to our site. We also have a bar that makes the browser reach a specific part of the page and a login and sign-up link all that to make it easier for the user. Figure 1.5 below represents the homepage :

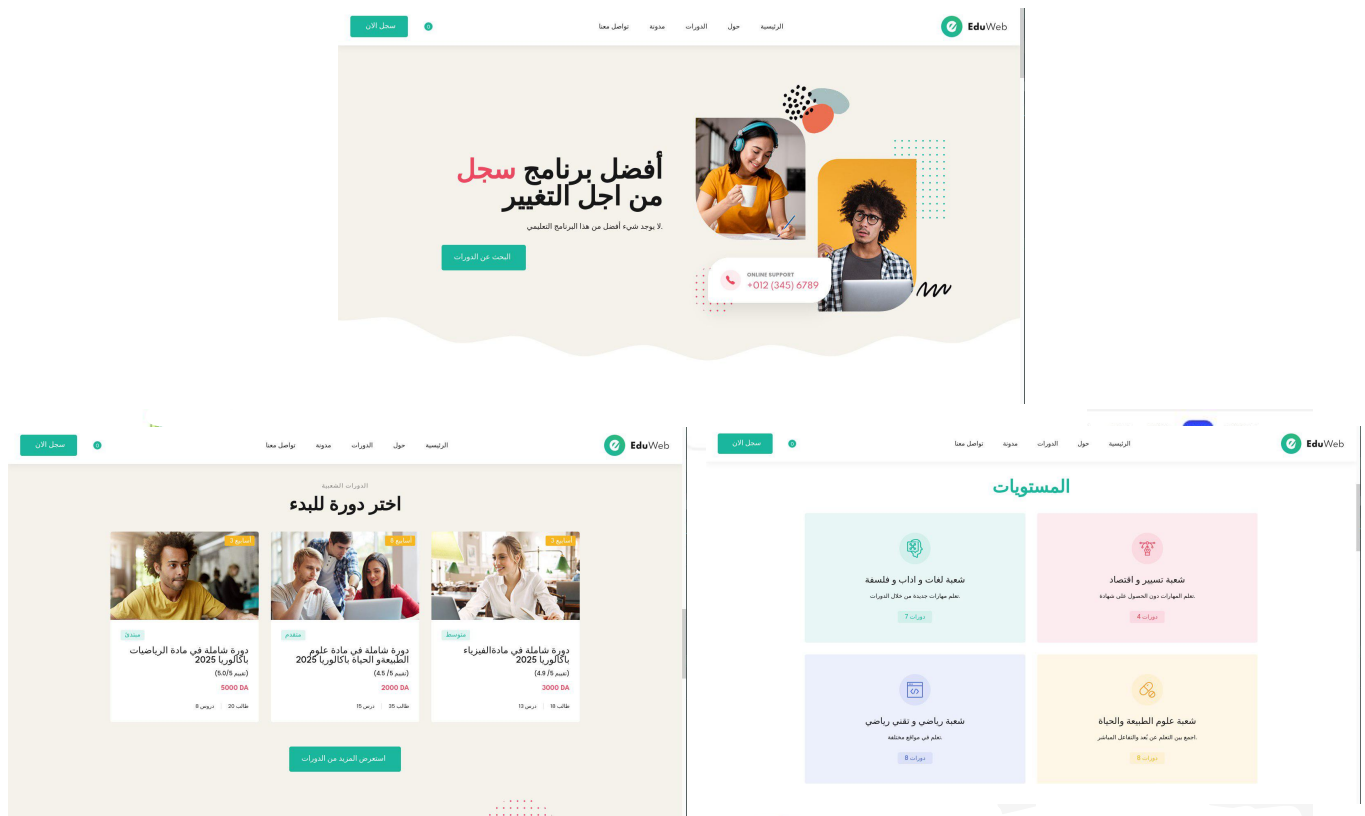


Figure 1.5 : Home Page Capture.

### 3.15 Login & sign-up Page

If the visitor does not have an account clicking on click on the login link to show a page that must be filled out by the information required to open an account (if he has an account to be logged in to his account), To be added to our website database you must click on the link sent in the email. Figure 1.6 below represents the login & sign-up Page:

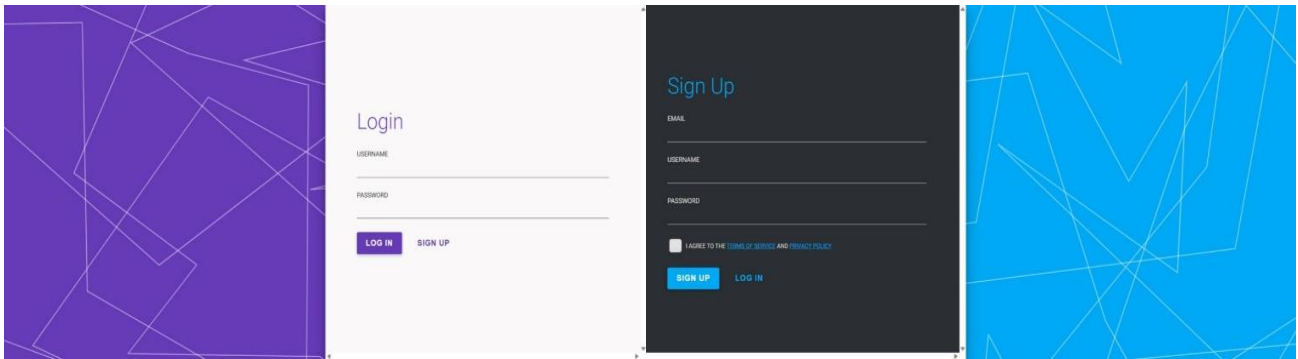


Figure 1.6 : Login & Sign-up Page

### 3.16 Teacher Profile Page

On the teacher profile page, students can find comprehensive information about the teacher, including: teacher's background, including their certificates and achievements in the field of education. Contacts such as email or social media handles. A list of courses offered by the teacher, along with descriptions and pricing information. Teaching Schedule. Additionally, students can follow the teacher, to stay updated on new courses and announcements. This page provides students with a detailed overview of the teacher's expertise and offerings, helping them make informed decisions about their learning journey. Figure 1.7 represents the teacher profile page.

### 3.17 Courses Page

On this page, we offer a variety of courses organized by year and field of study which are links we find in the home page. Here, students can access both paid and free modules, with prices displayed above the video thumbnails. To facilitate navigation, we've included a module bar that allows users to filter courses by specific modules. Additionally, a search bar is available to help students quickly find the courses they're looking for. For added convenience, we've included a "Home" link that allows users to easily return to the

homepage where they can explore all available content. The Figure 1.8 represents the courses page.



[الرئيسية](#)
[الدورات](#)
[برنامجي](#)

**الاسم: بن عثمان سيد علي**

**استاذ الرياضيات**

مهتم بالرياضيات ومكرس للتعليم، أنا مدرس في الثانوية ملتزم بإلهام وتوجيه طلابي في استكشاف العالم الرائع من حولهم. بنهج تعليمي ديناميكي وتفاعلي، أسعى لجعل المفاهيم العلمية متاحة ومثيرة. هدفني هو غرس المعرفة السليمة والفهم العميق للرياضيات والتفكير النقدي للطلاب، مما يجعلهم مستعدين لمواجهة تحديات العالم الحديث. معاً، نستكشف، نتعلم، ونفرض حياً دائماً للعلم والتعلم.

الإنشراح 1000 حج



الخبرة : 5 سنوات

البريد الإلكتروني : BenOthman@gmail.com

الشهادات : شهادة المقتورة في الرياضيات

المنطقة : النجيلة

## جدول التوقيت

مادة الرياضيات

| الوقت | الأحد                       | الاثنين                    | الثلاثاء                   | الأربعاء | الخميس                      | الجمعة                    | السبت                      |
|-------|-----------------------------|----------------------------|----------------------------|----------|-----------------------------|---------------------------|----------------------------|
| 08H   |                             |                            |                            |          |                             | تكنولوجيا<br>رياضيات      | تكنولوجيا<br>رياضيات       |
| 10H   |                             |                            |                            |          |                             | تكنولوجيا<br>رياضيات      | تكنولوجيا<br>رياضيات       |
| 12H   |                             |                            |                            |          |                             | تكنولوجيا<br>رياضيات      |                            |
| 14H   |                             |                            |                            |          |                             |                           | تكنولوجيا<br>تفكير رياضي   |
| 16H   |                             |                            | سنة أولى<br>علوم تجريبية   |          |                             | تكنولوجيا<br>علوم تجريبية | تكنولوجيا<br>تفكير رياضي   |
| 18H   | سنة الثانية<br>علوم تجريبية | سنة الثانية<br>تفكير رياضي | سنة الثانية<br>تفكير رياضي |          | سنة الثانية<br>علوم تجريبية |                           |                            |
| 20H   |                             | سنة الثانية<br>رياضيات     |                            |          | سنة الثانية<br>رياضيات      |                           | سنة الثانية<br>تفكير رياضي |

### الدورات

DA3000

القيم المضبوطة و القيم المقترنة

DA3000

القيم المضبوطة و القيم المقترنة

DA3000

القيم المضبوطة و القيم المقترنة

DA3000

القيم المضبوطة و القيم المقترنة

مستور

رسل

Figure 1.7 : Teacher profile page.



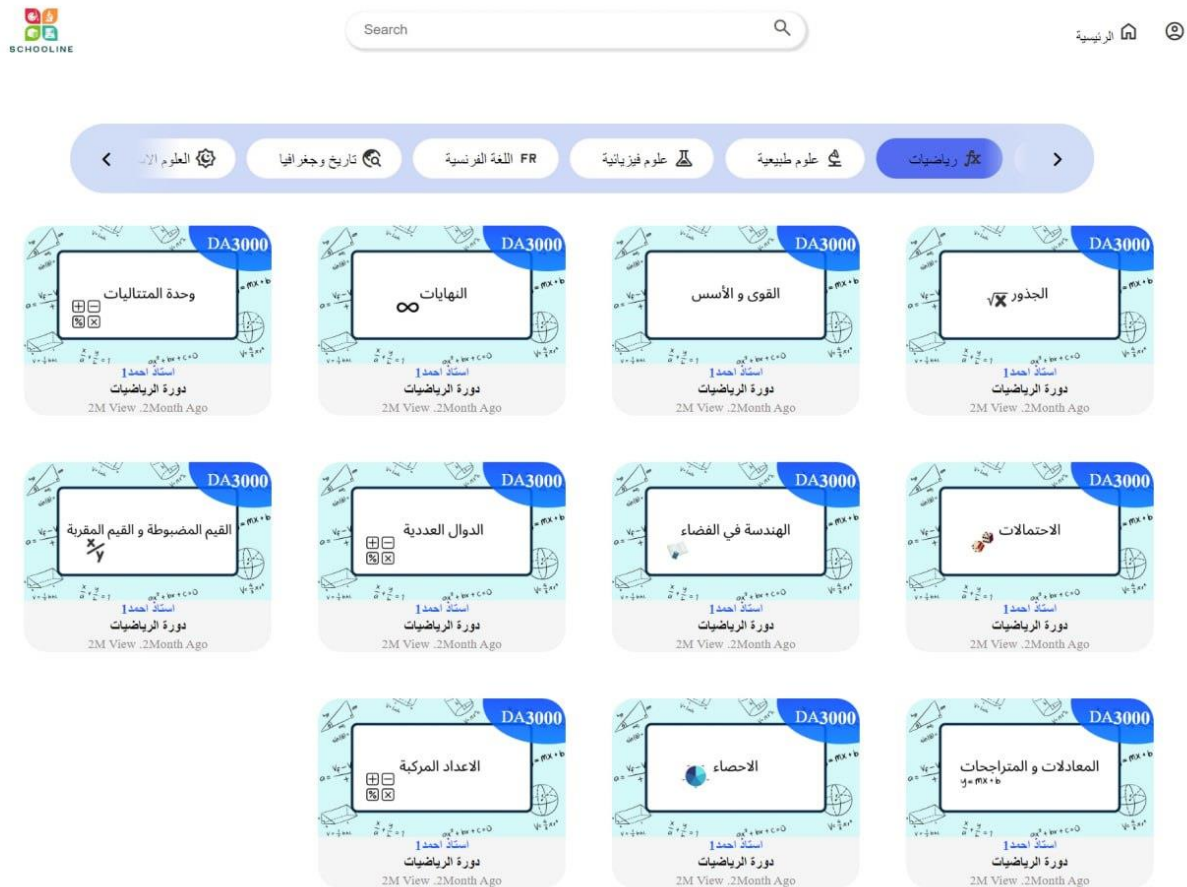


Figure 1.8 : Courses page.

### 3.18 Video Page

When a student clicks on a video thumbnail on the courses page, they are directed to the video page. Here, they can access the entire playlist of the selected course, which they have already paid for. Beneath the video, students can find like and dislike buttons, as well as a follow button to visit the teacher's page. Additionally, the teacher may upload documents for students, and students can leave comments, which only the teacher can respond to. Figure 1.9 represents the video page.

### 3.19 All Teachers Page

On the homepage, where a list of top teachers is displayed, users can click on the "More" button to access the "All Teachers" page. Here, students can find a comprehensive list of all teachers on the site. Students can click on the teacher's image to visit their page and access



**Figure 1.9 : Video Page.**

تاريخ و جغرافيا

رياضيات

علوم فيزيائية

علوم طبيعية

لغة عربية

لغة فرنسية

هندسة كهربائية

لغة انجليزية

فلسفة



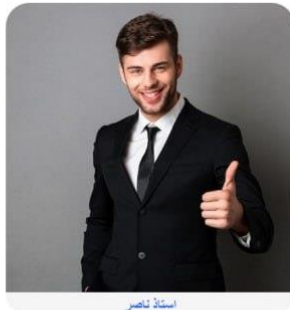
استاذ عبد الكريم



استاذ كمال



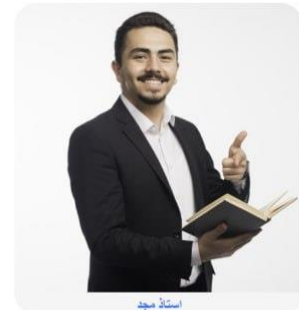
استاذ احمد



استاذ ناصر



استاذ كريم



استاذ ماجد

Figure 1.10 : All Teachers Page Capture.

### 3.20 Dashboard Page

the dashboard provides a comprehensive platform for students to view their profile information, communicate with others via messages, manage account settings, and interact with course-related data. It aims to offer a user-friendly experience for students navigating through different sections and performing various factions within the dashboard. Figure 1.11 represents the dashboard page.

#### 1/Navigation:

The navigation bar (<div class="navigation">) provides links to different sections of the dashboard:

**Profile (data-target="profile"):** Likely displays information about the student's profile, such as personal details, courses enrolled, and progress, following teachers, balance.

**Messages (data-target="messages"):** Provides access to a messaging system where students can communicate with the teach team to accept their payment and if there is a problem.

**Settings (data-target="settings"):** Allows students to manage account settings like profile information, security settings, and social media integration.

**Log Out (data-target="sign-out"):** Logs the user out of the account.

## **2/Profile Section:**

The profile section (<div id="profile" class="section">) seems to display various aspects related to the student's profile:

**Cards:** These cards display key metrics such as the number of courses, comments, and balance.

**Recent Courses:** Shows a list of recent courses with details like course name, price, teacher, and status (finished, in progress, not started).

**Recent Teachers:** Lists following teachers with their names, subjects they teach, and possibly their profile pictures.

## **3/Messages Section:**

The messages section (<div id="messages" class="section">) simulates a chat interface:

**Message Header:** Displays the teach team name and profile picture with whom the teach team is chatting.

**Chat Inbox:** Shows the conversation history, including incoming and outgoing messages, along with timestamps.

**Input Field:** Provides a text input area for sending new messages. There's also an option to upload images (<input type="file" id="image-upload">).

## **4/Settings Section:**

The settings section (<div id="settings" class="section">) allows students to manage their account settings:

**General Info:** Lets students update their first name, last name, and email address.

**Security Info:** Provides options to change the password, enable/disable two-factor authentication, and view logged-in devices.

**Social Info:** Allows students to link their social media accounts (Twitter, Facebook, LinkedIn, YouTube) to their dashboard.

### 5/logout section:

Logs the user out of the account and confirms their logout.

The screenshot displays the SchoolLine dashboard interface. On the left is a blue sidebar with navigation icons for Home (حساب), Messages (رسائل), Settings (اعدادات), and Logout (تسجيل الخروج). The main content area is white and contains several sections:

- General Information (معلومات عامة):** Includes a search bar at the top right. Below it, the 'Name' (اسم المستخدم) field contains 't' with a 'Change' button. The 'Email' field contains 'bensaadaterek05@gmail.com' with a 'Change' button.
- Security Info:** Titled 'Security Information About Your Account'. It features a 'Password' field with 'new Password' and a 'Change' button. Below this, it states 'Last Change On April 29, 2024, 4:56 p.m.'.
- Two-Factor Authentication:** A toggle switch is currently turned on (blue).
- Devices:** A section titled 'Check The Login Devices List' with a 'Devices' button.
- Social Info:** Titled 'Social Media Information', it includes input fields for 'Twitter Username', 'Facebook Username', 'LinkedIn Username', and 'YouTube Username', each with a 'Change' button.

**Figure 1.11** : Dashboard Page.

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## 4 CONCLUSION AND FUTURE WORKS

In conclusion, the "Schooline" project aims to revolutionize education in private schools through digital transformation. By addressing funding inadequacy and optimizing the educational ecosystem, "Schooline" provides affordable courses in various academic streams and additional subjects like language learning, design, and programming. Through an educational interface, teacher and course evaluation system, and the transition to digital teaching methods, "Schooline" seeks to enhance teaching quality, save time and effort for teachers, and create a productive and supportive learning environment for students. Looking ahead, "Schooline" will continue to evolve, further improving the overall quality of education in private schools and ensuring that students and teachers have access to the best possible learning and teaching experience.

For future website development, the following actions can be taken:

1. **Diverse Payment Options:** Implementing various electronic payment methods to facilitate seamless payment transactions.
2. **Academic Communication Platform:** Creating a platform for students to communicate, exchange knowledge, and share ideas.
3. **Enhanced User Interface and Experience:** Improving the website's user interface and experience to provide a more convenient and user-friendly platform.
4. **Personalized Learning Experience:** Utilizing data analytics to personalize the learning experience, meet individual student needs, and enhance security by implementing robust data protection measures.
5. **Live Courses:** Introducing live course options to provide real-time interaction between students and teachers, enhancing the overall learning experience.
6. **Adding new modules in the diverse page.**
7. **Adding sign language interpretation for videos** to make the platform more accessible to users with hearing impairments.
8. **Implementing a "VIP Student" feature** where the best-performing student receives a VIP card, allowing them to access the platform and study for free for a duration of three months.
9. **Bookmark Functionality:** Allow students to bookmark their favorite lessons for quick access later.
10. **Develop a mobile application** for the "Schooline" website.

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## 5 REFERENCES

W3schools: <https://www.w3schools.com/>.

Django : <https://www.djangoproject.com/>. & <https://www.dj4e.com/>.

Python: <https://www.python.org/>.

El zero-web school: <https://elzero.org/study/html-2021-study-plan/>.

Canva: <https://www.canva.com/>.

UML: <https://app.diagrams.net/>.

## 6 EACH MEMBER'S WORK

### Front end :

RABHI ADEM AHMED:Main page/Dashboard page/Rapport.

EL HADEF HAMZA RIADH:Main page/Logo/Video page/Teacher profile page.

BOUALAMALLAH DJAMEL EDDINE:Main page/Video page/Presentation.

BOURADA AMINE:Courses page.

BENBACHIR ABDERREZAKE:Teacher page.

### Backend:

BENBACHIR ABDERREZAKE:Linking the rest of pages and making the web site dynamic / fixing the data base structure / notification system.

BOURADA AMINE:Login and users system / notificcations system /linking teacher profile page/dashboard.